

PH 214 HW 2

① a) General form:

$$\vec{E} = (E_{0x} e^{i\phi_x} \hat{x} + E_{0y} e^{i\phi_y} \hat{y}) e^{i(kz - \omega t)}$$

• EM wave travelling in $\hat{z}$

↳ phase: $(kz - \omega t)$

• Magnitude $\|\vec{E}\| = 400 \text{ N/C}$

$$\hookrightarrow \|\vec{E}\| = \sqrt{\vec{E} \cdot \vec{E}^*}$$

$$\vec{E}^* = (E_{0x} e^{-i\phi_x} \hat{x} + E_{0y} e^{-i\phi_y} \hat{y}) e^{-i(kz - \omega t)}$$

$$\vec{E} \cdot \vec{E}^* = \left[(E_{0x} e^{i\phi_x} \hat{x} + E_{0y} e^{i\phi_y} \hat{y}) \cdot (E_{0x} e^{-i\phi_x} \hat{x} + E_{0y} e^{-i\phi_y} \hat{y}) \right] e^{i(kz - \omega t)} e^{-i(kz - \omega t)}$$

$\hat{x} \cdot \hat{x} \text{ and } \hat{y} \cdot \hat{y} = 1 \text{ but } \hat{x} \cdot \hat{y} \text{ and } \hat{y} \cdot \hat{x} = 0$

$e^0 = 1$

$$= E_{0x}^2 e^0 + E_{0y}^2 e^0$$

$$\Rightarrow \|\vec{E}\| = \sqrt{E_{0x}^2 + E_{0y}^2}$$

• Frequency $f = 1 \text{ Hz}$

$$\hookrightarrow f = \frac{1}{T}$$

$$v = \omega/k = c$$

$$\omega = 2\pi f$$

$$\Rightarrow \frac{2\pi f}{k} = c \Rightarrow k = \frac{2\pi f}{c}$$

• Elliptical polarization

↳ $E_{0x} \neq E_{0y}$ & $|\epsilon| \neq 0$ or $E_{0x} = E_{0y}$ & $|\epsilon| \neq \pi/2, 3\pi/2$

Therefore, picking $E_{0x} \neq E_{0y}$ will allow for any phase difference

• Let $E_{0x} = 300 \text{ N/C}$

$$\Rightarrow 400^2 = 300^2 + E_{0y}^2 \Rightarrow E_{0y} = 100\sqrt{7} \text{ N/C}$$

• Let $\phi_x = 0$, $\phi_y = \pi/2$ so $|\epsilon| = \pi/2 \neq 0$

$$\omega = 2\pi(1 \text{ Hz}) = 2\pi, \quad k = \frac{2\pi(1 \text{ Hz})}{c} = \frac{2\pi}{c}$$

$$\boxed{\vec{E} = (300 \hat{x} + 100\sqrt{7} e^{i\frac{\pi}{2}} \hat{y}) e^{i(\frac{2\pi}{c}z - 2\pi t)}}$$

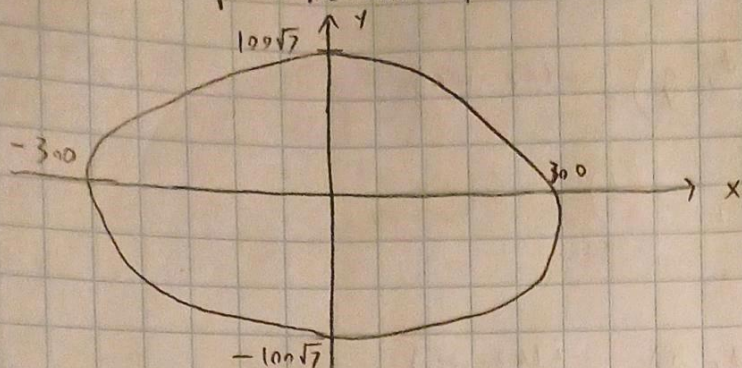
b) For $z=0$:

$$E_x = \text{Re} \{ 300 e^{i(-2\pi t)} \} = 300 \text{Re} \{ \cos(-2\pi t) + i \sin(-2\pi t) \} = 300 \cos(2\pi t)$$

$$E_y = \text{Re} \{ 100\sqrt{7} e^{i(\frac{\pi}{2} - 2\pi t)} \} = 100\sqrt{7} \text{Re} \{ \cos(\frac{\pi}{2} - 2\pi t) + i \sin(\frac{\pi}{2} - 2\pi t) \}$$

$$= 100\sqrt{7} \cos(\frac{\pi}{2} - 2\pi t)$$

Sketching the parametric for $t \in [0, 1]$:



$\alpha + \beta = 0.$

② Let $\vec{E} = E_x \hat{x} + E_y \hat{y} + E_z \hat{z}$

Plug into $\nabla^2 \vec{E} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2}$

$\Rightarrow \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (E_x \hat{x} + E_y \hat{y} + E_z \hat{z}) = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} (E_x \hat{x} + E_y \hat{y} + E_z \hat{z})$

$\text{In } x:$ $\frac{\partial^2 E_x}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 E_x}{\partial t^2}$	$\text{In } y:$ $\frac{\partial^2 E_y}{\partial y^2} = \frac{1}{c^2} \frac{\partial^2 E_y}{\partial t^2}$	$\text{In } z:$ $\frac{\partial^2 E_z}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 E_z}{\partial t^2}$
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$\hookrightarrow$ Thus the components of E are also waves.

③ $\vec{E} = (E_{0x} e^{i\phi_{xE}} \hat{x} + E_{0y} e^{i\phi_{yE}} \hat{y}) e^{i(kz - \omega t)}$

$\vec{B} = (B_{0x} e^{i\phi_{xB}} \hat{x} + B_{0y} e^{i\phi_{yB}} \hat{y}) e^{i(kz - \omega t)}$

a) • The phase $(kz - \omega t)$

$\hookrightarrow k$ (wavenumber) defines how spread out the wave is in space

$\hookrightarrow \omega$ (angular frequency) defines how spread out the wave in time (how often it oscillates)

$\hookrightarrow z$ (coordinate) defines direction of propagation

$\hookrightarrow t$ (time) defines time elapsed

• Magnitudes

$\hookrightarrow x$ and y components of the Electric field (E_{0x}, E_{0y})

$\hookrightarrow x$ and y components of the Magnetic field (B_{0x}, B_{0y})

• Phase relationship between the components of the $\vec{E}$ & $\vec{B}$ field

$\hookrightarrow \phi_{xE}, \phi_{yE}, \phi_{xB}, \phi_{yB}$

b) Let

$$\vec{E} = E_x \hat{x} + E_y \hat{y} + E_z \hat{z}$$

$$\vec{B} = B_x \hat{x} + B_y \hat{y} + B_z \hat{z}$$

Where $E_x, E_y, E_z, B_x, B_y, B_z$ respectively are of the form:

$$E_x = E_{0x} e^{i(kz - \omega t + \phi_{xE})}$$

$$1) \vec{\nabla} \cdot \vec{E} = 0$$

$$\left(\frac{\partial}{\partial x} E_x + \frac{\partial}{\partial y} E_y + \frac{\partial}{\partial z} E_z \right) = 0$$

$$\hookrightarrow \frac{\partial}{\partial x} E_x = \frac{\partial}{\partial y} E_y = 0 \quad \text{Since they are independent of } x \text{ and } y.$$

Thus,

$$\frac{\partial}{\partial z} E_z = \frac{\partial}{\partial z} \left(E_{0z} e^{i(kz - \omega t + \phi_{zE})} \right) = 0$$

$$\Rightarrow i k E_{0z} e^{i(kz - \omega t + \phi_{zE})} = 0$$

Since the exponential is never zero, E_{0z} must be zero.

$$2) \vec{\nabla} \cdot \vec{B} = 0$$

Similarly to (1), we can see that B_{0z} is also zero.

Note: In both (1) and (2) the lack of x & y dependence shows that the $\vec{E}$ and $\vec{B}$ fields are uniform at a fixed z and time.

$$3) \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & 0 \end{vmatrix} = -\frac{\partial}{\partial z} E_y \hat{x} + \frac{\partial}{\partial z} E_x \hat{y} + \left(\frac{\partial}{\partial x} E_y - \frac{\partial}{\partial y} E_x \right) \hat{z}$$

$$= -\frac{\partial}{\partial z} \left(E_{0y} e^{i(kz - \omega t + \phi_{yE})} \right) \hat{x} + \frac{\partial}{\partial z} \left(E_{0x} e^{i(kz - \omega t + \phi_{xE})} \right) \hat{y}$$

$$= -i k E_{0y} e^{i(kz - \omega t + \phi_{yE})} \hat{x} + i k E_{0x} e^{i(kz - \omega t + \phi_{xE})} \hat{y}$$

$$= i k e^{i(kz - \omega t)} \left[-E_{0y} e^{i\phi_{yE}} \hat{x} + E_{0x} e^{i\phi_{xE}} \hat{y} \right]$$

$$-\frac{\partial}{\partial t} (B_x \hat{x} + B_y \hat{y}) = - \left[\frac{\partial}{\partial t} (B_{0x} e^{i(kz - \omega t + \phi_{xB})}) \hat{x} + \frac{\partial}{\partial t} (B_{0y} e^{i(kz - \omega t + \phi_{yB})}) \hat{y} \right]$$

$$= i \omega B_{0x} e^{i(kz - \omega t + \phi_{xB})} \hat{x} + i \omega B_{0y} e^{i(kz - \omega t + \phi_{yB})} \hat{y}$$

$$= i \omega e^{i(kz - \omega t)} \left[B_{0x} e^{i\phi_{xB}} \hat{x} + B_{0y} e^{i\phi_{yB}} \hat{y} \right]$$

Setting the terms equal:

$$\underline{x}: -\kappa E_{0y} e^{i\phi_{1E}} = \omega B_{0x} e^{i\phi_{xB}}$$

$$\Rightarrow E_{0y} e^{i\pi} e^{i\phi_{1E}} = \frac{\omega}{\kappa} B_{0x} e^{i\phi_{xB}}$$

Thus, the magnitude & phase are equal:

$$\cdot \boxed{E_{0y} = \frac{\omega}{\kappa} B_{0x}} \quad \cdot \boxed{\phi_{1E} + \pi = \phi_{xB}}$$

$$\underline{y}: \kappa E_{0x} e^{i\phi_{xE}} = \omega B_{0y} e^{i\phi_{yB}}$$

$$\Rightarrow E_{0x} e^{i\phi_{xE}} = \frac{\omega}{\kappa} B_{0y} e^{i\phi_{yB}}$$

$$\cdot \boxed{E_{0x} = \frac{\omega}{\kappa} B_{0y}} \quad \cdot \boxed{\phi_{xE} = \phi_{yB}}$$

$$4) \vec{\nabla} \times \vec{B} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_x & B_y & 0 \end{vmatrix} = -\frac{\partial}{\partial z} B_y \hat{x} + \frac{\partial}{\partial z} B_x \hat{y} + \left(\frac{\partial}{\partial x} B_y - \frac{\partial}{\partial y} B_x \right) \hat{z}$$

$$= -\frac{\partial}{\partial z} (B_{0y} e^{i(kx - \omega t + \phi_{yB})}) \hat{x} + \frac{\partial}{\partial z} (B_{0x} e^{i(kx - \omega t + \phi_{xB})}) \hat{y}$$

$$= i\kappa e^{i(kx - \omega t)} [-B_{0y} e^{i\phi_{yB}} \hat{x} + B_{0x} e^{i\phi_{xB}} \hat{y}]$$

$$\frac{1}{c^2} \frac{\partial}{\partial t} (E_x \hat{x} + E_y \hat{y}) = \frac{1}{c^2} \left[\frac{\partial}{\partial t} (E_{0x} e^{i(kx - \omega t + \phi_{xE})}) \hat{x} + \frac{\partial}{\partial t} (E_{0y} e^{i(kx - \omega t + \phi_{yE})}) \hat{y} \right]$$

$$= \frac{1}{c^2} [-i\omega E_{0x} e^{i(kx - \omega t + \phi_{xE})} \hat{x} - i\omega E_{0y} e^{i(kx - \omega t + \phi_{yE})} \hat{y}]$$

$$= -\frac{1}{c^2} i\omega e^{i(kx - \omega t)} [E_{0x} e^{i\phi_{xE}} \hat{x} + E_{0y} e^{i\phi_{yE}} \hat{y}]$$

$$\underline{x}: -\kappa B_{0y} e^{i\phi_{yB}} = -\frac{1}{c^2} \omega E_{0x} e^{i\phi_{xE}}$$

$$\Rightarrow B_{0y} e^{i\phi_{yB}} = \frac{1}{c^2} \frac{\omega}{\kappa} E_{0x} e^{i\phi_{xE}}$$

$$\Rightarrow \boxed{B_{0y} = \frac{1}{c} E_{0x}, \quad \phi_{yB} = \phi_{xE}}$$

$$\underline{y}: \kappa B_{0x} e^{i\phi_{xB}} = -\frac{1}{c^2} \omega E_{0y} e^{i\phi_{1E}}$$

$$\Rightarrow B_{0x} e^{i\phi_{xB}} = \frac{1}{c^2} \frac{\omega}{\kappa} E_{0y} e^{i\pi} e^{i\phi_{1E}}$$

$$\Rightarrow \boxed{B_{0x} = \frac{1}{c} E_{0y}, \quad \phi_{xB} = \phi_{1E} + \pi}$$

$$(4) \vec{E} = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}, \quad \vec{k} = k_x \hat{x} + k_y \hat{y} + k_z \hat{z}, \quad \vec{r} = x \hat{x} + y \hat{y} + z \hat{z}$$

Recall: where ϕ is a scalar field & $\vec{A}$ is a vector field

$$\hookrightarrow \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + \vec{A} \cdot \vec{\nabla} \phi$$

$$\hookrightarrow \vec{\nabla} \times (\phi \vec{A}) = \vec{\nabla} \phi \times \vec{A} + \phi (\vec{\nabla} \times \vec{A})$$

$$\vec{\nabla} \cdot (\vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}) = e^{i(\vec{k} \cdot \vec{r} - \omega t)} \left(\vec{\nabla} \cdot \vec{E}_0 \right) + \vec{E}_0 \cdot (\vec{\nabla} e^{i(\vec{k} \cdot \vec{r} - \omega t)})$$

$\hookrightarrow$ Gauss's Law

$$\vec{\nabla} (e^{i(k_x x + k_y y + k_z z - \omega t)}) = i k_x e^{i(k_x x + k_y y + k_z z - \omega t)} \hat{x} \\ + i k_y e^{i(k_x x + k_y y + k_z z - \omega t)} \hat{y} \\ + i k_z e^{i(k_x x + k_y y + k_z z - \omega t)} \hat{z}$$

$$= i \vec{k} e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\Rightarrow \vec{\nabla} \cdot \vec{E} = (\vec{E}_0 \cdot i \vec{k}) e^{i(\vec{k} \cdot \vec{r} - \omega t)} = \left[(i \vec{k} \cdot \vec{E}_0) e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right]$$

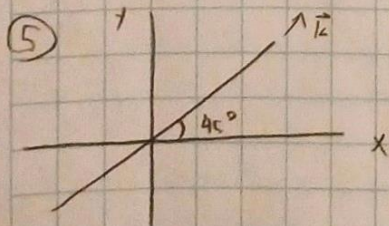
$$\vec{\nabla} \times (\vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}) = \vec{\nabla} (e^{i(\vec{k} \cdot \vec{r} - \omega t)}) \times \vec{E}_0 + e^{i(\vec{k} \cdot \vec{r} - \omega t)} (\vec{\nabla} \times \vec{E}_0)$$

$\vec{E}_0$ is constant so $\vec{\nabla} \times \vec{E}_0 = 0$

From above,

$$\vec{\nabla} (e^{i(\vec{k} \cdot \vec{r} - \omega t)}) = i \vec{k} e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\Rightarrow \vec{\nabla} \times \vec{E} = i \vec{k} e^{i(\vec{k} \cdot \vec{r} - \omega t)} \times \vec{E}_0 = \left[i \vec{k} \times \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right]$$



Direction: $\langle 1, 1, 0 \rangle \Rightarrow \|\langle 1, 1, 0 \rangle\| = \sqrt{2}$

$$\Rightarrow \hat{k} = \frac{1}{\sqrt{2}} \langle 1, 1, 0 \rangle$$

$$\left[\hat{k} = \frac{1}{\sqrt{2}} (\hat{x} + \hat{y}) \right]$$

Choosing $|\vec{E}| = \hat{z}$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow i \vec{k} \times \vec{E} = i \omega \vec{B} \Rightarrow \hat{B} = \hat{k} \times \hat{E}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{vmatrix} = \frac{1}{\sqrt{2}} \hat{x} - \frac{1}{\sqrt{2}} \hat{y} \Rightarrow \left[\hat{B} = \frac{1}{\sqrt{2}} (\hat{x} - \hat{y}) \right]$$

Check: $\hat{E} \times \hat{B} = \hat{k}$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{vmatrix} = \frac{1}{\sqrt{2}} \hat{x} + \frac{1}{\sqrt{2}} \hat{y} = \frac{1}{\sqrt{2}} (\hat{x} + \hat{y}) = \hat{k} //$$